

EE 526:Power Semiconductor Devices

Course Code : EE 526

Course Name :Power Semiconductor Devices

L-T-P-C : 3-0-0-3

Approval: 14th senate meeting

Prerequisite : IC160-Applied Electronics or equivalent, EE311-Device electronics for integrated circuit or equivalent

Intended for :BTech Third and Final year/M.Tech./MS/PhD

Distribution :Elective for third and final year Electrical Engineering, M.Tech. Power Electronics and Drives (PED), M.Tech VLSI, MS, PhD

Approval: 14th Senate

Course Contents

- **Introduction and Emergence of power semiconductor devices:** Overview of energy intensive civilization and its growth, impact on sustainable world, circuit requirements from power devices, power devices requirements from materials. [2 Lectures]
- **Materials Properties and Transport Physics:** Basics of semiconductor physics, polarization, quantum structures, bandgap narrowing, impact ionization, resistivity, recombination, scattering. [6 Lectures]
- **Semiconductor Junctions:** Review of p-n junction and metal-semiconductor junction, heterojunction, insulator- semiconductor junction, Zener and avalanche breakdown, design of breakdown voltage and edge terminations, parasitic circuit elements in rectifiers. [4 Lectures]
- **Power Diodes and rectifiers:** Power Schottky rectifiers, forward conduction and reverse blocking, device capacitance and thermal analysis, P-i-N rectifiers, switching performance. [4 Lectures]
- **5 BJT for Power Application:** Structure, operating principle, current gain, emitter current crowding, output and on-state and switching characteristics, Darlington configuration. [4 Lectures]
- **6 Power MOSFETs and HEMTs:** Ideal specific on-resistance, device structure and operation, characteristics, blocking voltage, VD-MOSFET, U-MOSFET, high frequency operation, switching characteristics, heterostructure, triangular potential well, 2DEG, charge control model, small-signal characteristics, power- frequency limit. [6 Lectures]
- **Thyristors:** Structure and operation, blocking and on-state and switching characteristics, Gate Turn-Off thyristor and triac structure and operations. [4 Lectures]
- **Insulated Gate Bipolar Transistor:** Structures, operation and output characteristics, equivalent circuit, blocking and on-state characteristics, current saturation model, power loss optimization, superjunction. [4 Lectures]
- **Widebandgap semiconductors and Advanced Technologies:** Properties and advantages of SiC and GaN, shielded technology, lateral and vertical device, enhancement mode device, reliability aspects, fabrication, homo-epitaxy, hetero-epitaxy, molecular beam epitaxy, metal organic chemical vapor deposition, packaging and thermal management, power IC, integration of devices with CMOS, oxide electronics. [8 Lectures]

Students will perform one project/practical work based on modeling, simulation with Technology Computer Aided Design (Synopsys Sentaurus/ Silvaco ATLAS) or fabrication for further improvement of devices performances as instructed.

Text books:

1. B. J. Baliga, **Fundamentals of Power Semiconductor Devices**, Springer, 2008.
2. Yung C Liang and Ganesh S Samudra, **Power Microelectronics: Device and Process Technologies**, World Scientific.

References:

1. B. Streetman and S. Banerjee, **Solid State Electronic Devices**, 7th Edition, 2006.
2. S.M. Sze, **Physics of Semiconductor Devices**, 2nd Edition, Wiley, 2008
3. M. Meneghini, G. Meneghesso, E. Zanoni, **Power GaN Devices: Materials, Applications and Reliability**, Springer, 2017.
4. Tore M. Undeland, Riobbbins, **Power electronics: converters, applications, and design**, Wiley, 2007.
5. R.S. Ramshaw, **Power Electronics Semiconductor Switches**, Champman & Hall, 2013.